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Research Paper

GCR: A Global Cable Routing Planner for Two-Dimensional Cluttered Environments

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Abstract

Planning collision-free motions for deformable linear objects (DLOs), such as cables, wires, and ropes, is difficult because the object state is defined by its complete shape rather than by endpoint positions alone. This paper presents the Global Cable Routing (GCR) planner for cable-like DLOs in two-dimensional cluttered workspaces. The cable is represented as an ordered polygonal chain, and the planner searches directly in the corresponding discretized configuration space. GCR combines bidirectional tree search, guided configuration sampling, configuration-level artificial potential field biasing, projected elastic relaxation, ordered edge validation, and shortcut refinement. Candidate configurations are biased toward the opposite tree and away from obstacles, then corrected by a reference-based relaxation step that penalizes unnecessary changes in segment vectors and discrete curvature. A candidate is inserted into the tree only if it is statically admissible and if the ordered transition from a feasible parent satisfies swept-envelope collision checking, endpoint-axis consistency, and endpoint feasibility constraints. This prevents the planner from accepting two individually valid cable shapes when the local deformation between them would collide with obstacles or violate endpoint correspondence. Benchmark results in representative planar environments and simulation-based dual-arm validation indicate that GCR can produce compact, collision-free, and transition-feasible cable-routing paths under the tested conditions.

Keywords: deformable linear objects, cable routing, motion planning, sampling-based planning, artificial potential fields, dual-arm manipulation.

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1. Introduction

Deformable linear objects (DLOs), such as cables, ropes, wires, hoses, and sutures, appear in many robotic applications, including wire-harness assembly, cable routing, medical suturing, and domestic manipulation. Despite their practical importance, robotic manipulation of DLOs remains difficult because the object shape is not described by a small set of rigid-body coordinates. Instead, the object may bend, stretch slightly, slide around obstacles, and form many admissible geometries under the same endpoint positions. As a result, DLO manipulation requires simultaneous reasoning about geometry, deformation, collision avoidance, and mechanical feasibility **????**.

A large part of previous research on DLO manipulation has focused on local shape control. In this setting, the robot deforms the object from an observed shape toward a desired one, often using visual feedback, model-based prediction, or an estimated deformation Jacobian. In our previous work, a featureless cable was represented by virtual feature points, and a Jacobian-based manipulation strategy was developed to relate dual-arm end-effector motion to the motion of sampled cable points **?**. This idea was later extended through intermediate cable profiles and optimization-based dual-arm planning for cable shape control **??**. These works show that point-based cable representations and optimization-based manipulation are useful for planar shaping tasks. However, they mainly address deformation toward a desired profile and do not solve the global routing problem in cluttered environments, where the cable must move through obstacle-constrained regions while maintaining collision-free transitions.

Global cable routing requires a different planning formulation. A planner must not only find collision-free cable configurations, but also ensure that the deformation between consecutive configurations is feasible. Two cable shapes may be individually collision-free, while the swept region between them intersects an obstacle. Therefore, the planning problem must be defined in the configuration space of the complete cable, and tree edges must be validated as local deformations rather than treated only as connections between nearby states.

Classical sampling-based motion planning methods, such as RRT, RRT-Connect, and RRT*, are attractive for high-dimensional problems because they avoid explicit construction of the complete configuration-space obstacle region **??**. However, direct application of these methods to DLOs is inefficient because random polygonal chains are usually mechanically invalid, excessively stretched, or collision-prone. Earlier DLO planners addressed this problem by sampling stable or minimum-energy shapes **?**, by characterizing equilibrium manifolds of elastic rods **?**, or by planning elastic-rod motion with contacts **?**. These approaches provide important theoretical foundations, but high-fidelity rod models and contact-rich planning can be computationally expensive for online routing in cluttered workspaces.

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This paper proposes the Global Cable Routing (GCR) planner for a cable-like DLO in a two-dimensional cluttered workspace. The cable is represented as an ordered sequence of points connected by fixed-length segments. The planner searches directly in the discretized cable configuration space and evaluates both static configuration feasibility and ordered transition feasibility. Local deformation is measured through segment-vector and discrete-curvature changes, allowing the planner to penalize unnecessary stretching, compression, rotation, and bending between neighboring configurations.

The proposed method combines bidirectional tree search, guided configuration sampling, configuration-level artificial potential field (APF) biasing, projected elastic relaxation, ordered parent selection, ordered tree connection, and shortcut refinement. The APF component biases raw samples toward the opposite tree and away from obstacles, while the tree structure preserves global exploration. The relaxation step locally corrects APF-biased samples by keeping them close to the sampled direction while preserving segment vectors and discrete curvature relative to a reference configuration. This reference-based regularization avoids driving the cable only toward a low-energy shape and instead keeps the deformation local with respect to the previously feasible configuration. A central part of the formulation is the ordered transition predicate. Since the cable is represented as an ordered sequence of points, reversing only one of two neighboring configurations changes the point-to-point correspondence used during deformation. This may change the swept region and therefore the collision result. For this reason, GCR validates each accepted edge using the ordered edge-feasibility predicate, which combines swept-envelope collision checking, endpoint-axis consistency, and optional robot-dependent endpoint feasibility. Thus, endpoint ordering is considered during parent selection, tree connection, and shortcut refinement, not only after a path has already been generated. The main contributions of this paper are as follows:

1. A global cable-routing formulation in the discretized configuration space of an ordered planar cable, including static admissibility, reference-based local deformation measures, transition feasibility, and endpoint correspondence.
2. A bidirectional sampling-based planner that combines guided sampling and configuration-level APF biasing to generate candidate cable configurations in cluttered workspaces.
3. A projected elastic relaxation step that regularizes APF-biased samples by preserving segment vectors and discrete curvature with respect to a reference configuration, reducing unnecessary local deformation during planning.
4. An ordered edge-validation strategy for parent selection, tree connection, and shortcut refinement. The strategy checks swept collision, endpoint-axis consistency, and robot-dependent endpoint feasibility through a unified ordered transition predicate.
5. A benchmark study and simulation-based dual-arm manipulation validation showing that the proposed planner produces compact, collision-free, and transition-feasible cable-routing paths under the tested conditions.

The remainder of the paper is organized as follows. Section **??** discusses related work on DLO modeling, manipulation, and obstacle-aware planning. Section **??** formulates the cable-routing problem. Section **??** presents the proposed GCR algorithm. Section **??** reports the benchmark and robotic validation results. Section **??** discusses the results, limitations, and future directions. Section **??** concludes the paper.

2. Related Work

Robotic manipulation of deformable linear objects (DLOs), such as cables, wires, ropes, rods, and sutures, has been studied in several contexts, including shape control, routing, insertion, transport, and manipulation in constrained environments **????**. Although the general field has grown rapidly, global planning for cable-like objects in cluttered environments remains less explored than local shape control or task-specific manipulation. The main difficulty is that a DLO is not sufficiently described by the positions of its endpoints. Its complete shape must be considered, and in obstacle-rich environments the transition between two shapes can be invalid even when both shapes are individually collision-free.

A large group of DLO manipulation methods focuses on local shape control. These methods usually assume that a desired shape or a sequence of desired shapes is already given, and the controller computes robot motions to reduce the shape error. Jacobian-based methods estimate a local relation between robot motion and DLO deformation, while learning-based methods approximate this relation from data **????**.

Our earlier work followed this local shape-control direction. In **?**, a planar cable was represented as a set of points, and an analytical Jacobian was derived to map the motion of two robot end-effectors to the displacement of the cable points while preserving the length constraint. This formulation was extended in **?**, where a featureless cable was detected and sampled using virtual feature points, and intermediate profiles were generated to guide the cable toward the desired shape in real experiments. A later extension addressed large and complex planar deformations by introducing an intermediary-shape generation strategy and an optimization-based co-manipulation controller **?**. These works showed that point-based DLO representations and intermediate shape generation are effective for planar shape control. However, they assume that the target shape or intermediate profiles are provided or generated for local deformation, and they do not solve the global routing problem in cluttered environments.

Thus, although local shape-control methods can deform and track DLO shapes, they are not sufficient when the cable must be routed through obstacle-constrained regions. In such cases, a planner must decide not only the final shape, but also the sequence of intermediate cable configurations and the feasibility of the deformation between consecutive configurations.

Several works address path planning for DLOs using physically motivated models. Early studies formulated DLO planning by sampling stable or minimum-energy configurations **?**. Bretl and McCarthy analyzed quasi-static manipulation of Kirchhoff elastic rods and showed that equilibrium configurations can be represented on a finite-dimensional manifold **?**. Roussel et al. extended this line of work to elastic rods with contacts **?**. Sintov et al. reduced online planning time by precomputing a roadmap of stable elastic-rod configurations for dual-arm manipulation **?**. These methods provide important theoretical foundations and physically meaningful representations. However, they often require equilibrium computation, accurate mechanical parameters, contact-rich simulation, or precomputed roadmaps. These requirements can limit their use as lightweight global routing planners, especially when many candidate cable configurations must be generated and checked.

Recent works have attempted to make DLO planning more deployable in constrained and cluttered environments. Yu et al. proposed a coarse-to-fine framework for dual-arm manipulation of DLOs with whole-body obstacle avoidance, where a simplified elastic model is used for global planning and a local controller compensates for model errors during execution **?**. Their later work extended this idea to three-dimensional constrained environments by considering the DLO, the robot arms, stable deformation, closed-chain constraints, overstretch prevention, and collision avoidance within a whole-body planning-control framework **?**. This line of work is important because it demonstrates that global DLO manipulation in cluttered environments can be achieved when planning is supported by closed-loop feedback. However, the global plan is still used mainly as a guidance path, and execution robustness depends on the controller compensating for modeling errors and simplifications in the planning model.

A similar planning-control direction was proposed by Aksoy and Wen **?**. Their framework uses an off-the-shelf global planner by approximating the DLO as a chain of rigid links connected by spherical joints. A real-time controller based on control barrier functions and position-based dynamics then enforces obstacle avoidance, stress limits, and DLO integrity. This approach is practical and attractive because it avoids designing a specialized DLO planner and can be deployed with standard planning tools. However, the global planner produces a coarse path for the approximated object, while fine deformation behavior, stress constraints, and modeling errors are handled mainly by the local controller. In contrast, the objective of GCR is to reason directly about the full planar cable configuration during planning and to validate each accepted planning edge as a local cable deformation.

Other recent approaches reduce the difficulty of constrained DLO planning by using hierarchical representations. Tang et al. proposed a framework that first generates a spatial path set satisfying homotopic constraints for DLO keypoints, then optimizes a temporal deformation sequence and tracks it using a neural model-predictive controller **?**. This formulation is effective for constrained manipulation because it reduces the search space before deformation optimization. At the same time, it introduces a multi-stage pipeline combining path-set generation, deformation optimization, and learned tracking. Such hierarchical approaches are powerful, but they do not directly address the specific problem considered here: constructing a compact sequence of full cable configurations in a planar cluttered workspace while explicitly validating each local swept deformation between consecutive configurations.

Cable routing has also been studied in task-specific settings. For example, fixture-based or assembly-oriented methods represent the task through cable features, fixtures, grasp actions, or local routing operations **???**. These methods are relevant to practical cable assembly and can be efficient when the task structure is known. However, they often rely on specific fixtures, predefined action primitives, or simplified representations of the cable state. Consequently, they do not generally solve the full-shape planning problem where the planner must decide how the entire cable centerline should move through a cluttered environment.

The literature shows a clear gap. Local controllers can deform and track DLO shapes but do not solve global routing. High-fidelity rod planners can represent deformation accurately but are often computationally demanding. Planning-control frameworks can handle complex environments, but the global planner usually provides a coarse guide while the local controller compensates for deformation errors. Task-specific cable-routing systems are effective for structured assembly problems but do not provide a general full-shape routing planner. Moreover, most existing approaches validate static configurations or track planned references, while less attention is given to explicit validation of the swept region generated by the local deformation between two neighboring cable configurations.

The proposed GCR planner addresses this gap for the planar case. It searches directly in the discretized cable configuration space, where each state is the full ordered cable centerline. Candidate configurations are guided by a configuration-level artificial potential field and corrected using local deformation regularization. More importantly, accepted edges are validated as ordered transitions: the swept envelope between consecutive configurations is checked, and endpoint correspondence is preserved during parent selection, tree connection, and shortcut refinement. Thus, GCR is positioned between lightweight geometric planners and high-fidelity physical DLO planners. It does not aim to replace full physical simulation or closed-loop execution control, but it provides a compact global routing layer that explicitly reasons about both static cable feasibility and transition-level deformation feasibility.

3. Problem Formulation

This section defines the planning problem solved by the implemented Global Cable Routing (GCR) benchmark. The deformable linear object (DLO) is represented as a planar polygonal chain. The objective is to find a finite sequence of cable configurations that connects a given start configuration to a given goal configuration while satisfying static collision constraints and transition-level deformation feasibility.

3.1. Workspace and Obstacles

Let

$$\mathscr{W} = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \subset \mathbb{R}^2 \tag{1}$$

be a bounded planar workspace. The obstacle set is denoted by

$$\mathscr{O}_{\text{set}} = \{O_1, O_2, \ldots, O_M\}, \tag{2}$$

where each obstacle is represented either as a polygon or as a circle. The total occupied region is

$$\mathscr{O} = \bigcup_{m=1}^{M} O_m, \tag{3}$$

and the collision-free workspace is

$$\mathscr{F} = \mathscr{W} \setminus \mathscr{O}. \tag{4}$$

The obstacle geometry is assumed to be known and fixed during planning. The implemented benchmark checks collisions against polygonal obstacle boundaries and circular obstacles directly. Therefore, collision validity is evaluated geometrically rather than through an occupancy-grid approximation.

3.2. Discrete Cable Representation

A cable configuration is represented by an ordered sequence of N planar nodes,

$$C = (p_1, p_2, \ldots, p_N), \qquad p_n = (x_n, y_n) \in \mathbb{R}^2, \qquad n = 1, \ldots, N. \tag{5}$$

The cable configuration is approximated by the polygonal chain connecting consecutive nodes. The k-th segment is

$$s_k(C) = [p_k, p_{k+1}], \qquad k = 1, \dots, N-1. \tag{6}$$

The ambient configuration space is therefore

$$\mathbf{X} = \mathbb{R}^{2N}. \tag{7}$$

The start and goal configurations are denoted by

$$C_{\text{start}}, C_{\text{goal}} \in \mathbf{X}. \tag{8}$$

Both configurations are validated before the benchmark is executed. In particular, they must have the same number of nodes, they must lie inside the workspace, their segment lengths must satisfy the admissible bounds, and their cable segments must be collision-free.

The nominal segment length used by the implementation is computed from the start configuration:

$$\ell_0 = \frac{1}{N-1} \sum_{k=1}^{N-1} \|p_{k+1}^{\text{start}} - p_k^{\text{start}}\|. \tag{9}$$

Figure **??** illustrates the workspace model used in the benchmark. The rectangular region defines the bounded workspace $\mathscr{W}$, the red regions represent the static obstacles $\mathscr{O}_{\text{set}}$, and the remaining area corresponds to the collision-free space $\mathscr{F}$.

The same figure also shows the start and goal cable configurations as ordered polygonal chains. The first and last nodes, p_1 and p_N, correspond to the robot end-effectors, while the intermediate nodes discretize the deformable cable geometry.

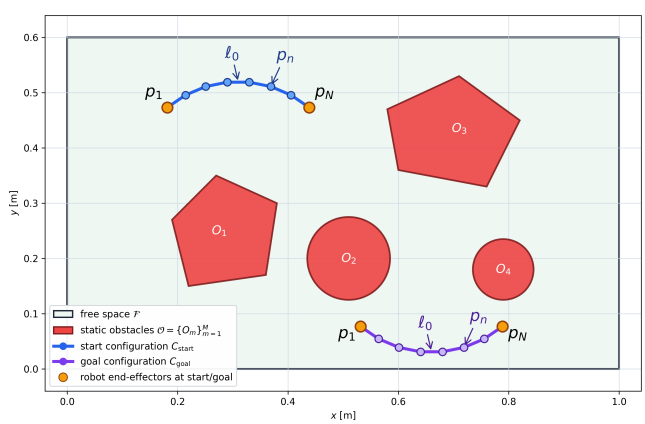


Figure 1: Problem formulation for global cable routing in a planar workspace with static obstacles.

3.3. Local Deformation Measures

The planner does not simulate the full dynamic behavior of the cable during search. Instead, it uses local geometric deformation measures to describe how the cable shape changes between two configurations. These measures are based on segment-vector variation and discrete-curvature variation. They are used both to regularize candidate configurations during relaxation and to evaluate the deformation cost of accepted planning edges.

For a cable configuration

$$C = (p_1, p_2, \dots, p_N), \tag{10}$$

the vector of the k-th cable segment is defined as

$$v_k(C) = p_{k+1} - p_k, \qquad k = 1, \dots, N-1. \tag{11}$$

This vector represents both the local length and the local orientation of the cable segment. Therefore, changes in $v_k(C)$ describe local stretching, compression, and rotation of the corresponding cable element.

Local bending is described by the second-order finite difference

$$b_k(C) = p_{k+2} - 2p_{k+1} + p_k, \qquad k = 1, \dots, N-2. \tag{12}$$

This quantity measures the local change of direction along the cable. Larger values correspond to sharper local curvature, while smaller values correspond to smoother local shapes.

Let

$$C^a = (p_1^a, p_2^a, \dots, p_N^a), \qquad C^b = (p_1^b, p_2^b, \dots, p_N^b) \tag{13}$$

be two cable configurations with the same number of nodes. The segment-vector change between them is

$$\Delta v_k(C^a, C^b) = v_k(C^a) - v_k(C^b), \qquad k = 1, \dots, N-1, \tag{14}$$

and the curvature change is

$$\Delta b_k(C^a, C^b) = b_k(C^a) - b_k(C^b), \qquad k = 1, \dots, N-2. \tag{15}$$

The local deformation energy between the two configurations is then defined as

$$E_{\text{def}}(C^a, C^b) = \lambda_s \sum_{k=1}^{N-1} \left\| \Delta v_k(C^a, C^b) \right\|^2 + \lambda_b \sum_{k=1}^{N-2} \left\| \Delta b_k(C^a, C^b) \right\|^2, \tag{16}$$

where $\lambda_s > 0$ and $\lambda_b > 0$ weight the preservation of local segment structure and local curvature, respectively.

A small value of $E_{\text{def}}(C^a, C^b)$ indicates that the transition preserves the local segment lengths, segment directions, and curvature of the cable. A large value indicates an abrupt local deformation, such as excessive stretching, compression, rotation, or bending. Therefore, E_{def} is used as the deformation penalty in edge-cost computation and path-quality evaluation.

The same measure is also used during elastic relaxation. In that case, one of the two configurations is the candidate configuration being optimized, and the other is a reference configuration C^{ref}. This reference-based form reflects the local nature of the planning problem. The objective is not only to obtain a smooth or equilibrium-like cable shape. Minimizing only an absolute stretching–bending energy may produce a low-energy configuration that is far from the previous feasible configuration. Such a change can introduce an unnecessarily large deformation along a single planning edge. In GCR, each edge represents a local manipulation step. Therefore, deformation is evaluated relative to a nearby feasible configuration, so that the planner preserves local cable structure while still allowing the shape to change when required for obstacle avoidance and tree expansion.

3.4. Admissible and Collision-Free Configurations

A cable configuration is considered statically admissible when it satisfies three conditions. First, all cable nodes must lie inside the workspace. Second, the deformation of each cable segment must remain bounded with respect to a reference configuration. Third, no cable segment may intersect the obstacle region.

Let C^{ref} denote the feasible reference configuration used to evaluate local deformation. Using the segment-vector deviation $\Delta v_k(C, C^{\text{ref}})$ defined in (**??**), the local change of a cable segment is measured in both length and direction. To prevent excessive stretching, compression, or local rotation, this segment-vector deviation is bounded as

$$\|\Delta v_k(C, C^{\mathrm{ref}})\| \le \delta_\ell, \qquad k = 1, \dots, N-1, \tag{17}$$

where $\delta_\ell > 0$ is the maximum allowed local segment-vector deviation. Since this bound is evaluated relative to C^{ref}, the admissible set is reference-dependent. We write

$$\begin{aligned} \mathscr{C}_{\mathrm{free}}(C^{\mathrm{ref}}) = \{C \in \mathbf{X} \mid\ & p_n \in \mathscr{W}, && n = 1, \dots, N, \\ & \|\Delta v_k(C, C^{\mathrm{ref}})\| \le \delta_\ell, && k = 1, \dots, N-1, \\ & s_k(C) \cap \mathscr{O} = \emptyset, && k = 1, \dots, N-1\} \end{aligned} \tag{18}$$

and use $\mathscr{C}_{\mathrm{free}}$ as shorthand when the reference configuration is clear from context.

This definition checks the complete polygonal representation of the cable, not only its discrete nodes. In particular, obstacle avoidance is verified for every segment

$$s_k(C) = [p_k, p_{k+1}],$$

which prevents collisions along the cable configuration between neighboring nodes.

To measure the safety margin between the cable and the obstacles, we use a signed obstacle distance. For a point $x \in \mathbb{R}^2$, this distance is defined as

$$d_{\mathscr{O}}(x) = \begin{cases} \mathrm{dist}(x, \partial\mathscr{O}), & x \notin \mathscr{O}, \\ 0, & x \in \partial\mathscr{O}, \\ -\,\mathrm{dist}(x, \partial\mathscr{O}), & x \in \mathscr{O}. \end{cases} \tag{19}$$

Thus, a positive value indicates that the point is outside the obstacle region, a zero value indicates contact with the obstacle boundary, and a negative value indicates penetration into an obstacle.

The clearance of a complete cable configuration is evaluated by sampling points along all cable segments and taking the smallest signed distance among them. For the q-th sample on segment $s_k(C)$, let

$$x_{kq}(C) = (1-\rho_q)p_k + \rho_q p_{k+1}, \qquad \rho_q = \frac{q}{Q}, \qquad q = 0, \dots, Q.$$

The configuration clearance is then

$$\mathrm{clear}(C) = \min_{\substack{k=1,\dots,N-1 \\ q=0,\dots,Q}} d_{\mathscr{O}}\left(x_{kq}(C)\right). \tag{20}$$

A positive clearance means that the sampled cable is separated from the obstacles. A zero clearance means that the cable touches an obstacle boundary. A negative clearance means that at least one sampled point lies inside an obstacle, and therefore the configuration is in collision.

3.5. Configuration Distance and Path Cost

The distance between two cable configurations is measured by the maximum point-wise displacement of corresponding cable nodes:

$$d(C_i, C_j) = \max_{1 \le n \le N} \left\| p_n^i - p_n^j \right\|. \tag{21}$$

This metric limits the largest motion of any discretized cable point during a local transition. It is therefore suitable for nearest-neighbor search, connection-radius tests, local edge evaluation, and path-cost computation.

A discrete cable path is represented as a finite sequence of configurations,

$$\mathscr{P} = \{C_0, C_1, \dots, C_K\}, \qquad C_0 = C_{\mathrm{start}}, \qquad C_K = C_{\mathrm{goal}}. \tag{22}$$

Each consecutive pair (C_i, C_{i+1}) defines one local deformation of the cable. The geometric length of the path is computed as

$$J_{\mathrm{geom}}(\mathscr{P}) = \sum_{i=0}^{K-1} d(C_i, C_{i+1}). \tag{23}$$

In addition to geometric displacement, the planner penalizes deformation between consecutive cable configurations. The deformation cost of the path is defined as

$$J_{\mathrm{def}}(\mathscr{P}) = \sum_{i=0}^{K-1} E_{\mathrm{def}}(C_{i+1}, C_i), \tag{24}$$

where E_{def} is the local deformation energy defined in (**??**). This term penalizes abrupt changes in segment vectors and discrete curvature along the path.

The complete path objective combines the geometric motion cost and the deformation cost:

$$J(\mathscr{P}) = J_{\mathrm{geom}}(\mathscr{P}) + \alpha_{\mathrm{def}} J_{\mathrm{def}}(\mathscr{P}), \tag{25}$$

where $\alpha_{\mathrm{def}} \ge 0$ is a weighting coefficient that controls the relative importance of deformation minimization.

The first term favors short transitions in configuration space, while the second term favors smooth and mechanically plausible cable deformation. Therefore, a low-cost path is not only short in terms of node displacement, but also avoids unnecessary stretching, compression, and bending between consecutive configurations.

3.6. Transition Feasibility

Static feasibility of individual cable configurations is not sufficient for cable manipulation. Even if two configurations are separately collision-free, the motion between them may pass through an obstacle. Therefore, every edge between two neighboring configurations must be checked for transition feasibility.

Let

$$C^a = (p_1^a, p_2^a, \dots, p_N^a), \qquad C^b = (p_1^b, p_2^b, \dots, p_N^b)$$

be two cable configurations. During the transition from C^a to C^b, the cable occupies a swept region in the workspace. In the planar case, this region is represented by the polygonal envelope formed by the two cable shapes:

$$\mathscr{E}_{ab} = \mathrm{Poly}\left(p_1^a, p_2^a, \dots, p_N^a, p_N^b, p_{N-1}^b, \dots, p_1^b\right). \tag{26}$$

The transition is considered deformation-feasible if this swept envelope does not intersect the obstacle region:

$$\mathscr{E}_{ab} \cap \mathscr{O} = \emptyset. \tag{27}$$

Accordingly, the binary transition-validity function is defined as

$$\mathrm{PathFree}(C^a, C^b) = \begin{cases} 1, & \mathscr{E}_{ab} \cap \mathscr{O} = \emptyset, \\ 0, & \text{otherwise}. \end{cases} \tag{28}$$

Thus, an accepted edge represents a collision-free local deformation between two cable configurations, not only two individually admissible configurations. The swept-envelope test ensures that the cable does not pass through an obstacle during the transition.

3.7. Ordered Transition Feasibility and Endpoint Correspondence

The transition feasibility condition introduced in Section **??** depends not only on the geometric shape occupied by two cable configurations, but also on the point-to-point correspondence assumed during the transition. This distinction is important because the cable is represented as an ordered sequence of nodes. The ordering determines which cable endpoint is associated with the first robot end-effector and which endpoint is associated with the second robot end-effector.

For a cable configuration

$$C = (p_1, p_2, \dots, p_N), \tag{29}$$

we define the reversal operator

$$R(C) = (p_N, p_{N-1}, \dots, p_1). \tag{30}$$

The configurations C and $R(C)$ describe the same static cable configuration. Therefore, for static collision checking, they are geometrically equivalent. However, they do not necessarily define the same transition when connected to another ordered configuration.

Let

$$C = (p_1, p_2, \dots, p_N), \qquad D = (q_1, q_2, \dots, q_N) \tag{31}$$

be two ordered cable configurations. The swept-envelope test used by $\mathrm{PathFree}(C,D)$ assumes that point p_n moves toward point q_n for all $n = 1, \dots, N$. Thus, the transition is defined by the ordered correspondence

$$p_n \longleftrightarrow q_n, \qquad n = 1, \dots, N. \tag{32}$$

If one of the two configurations is reversed, this correspondence changes. For example, the transition from $R(C)$ to D connects p_{N-n+1} to q_n, which may generate a different swept region from the one generated by the transition from C to D.

This motivates the following observation.

Lemma: Simultaneous Reversal Invariance. Let C and D be two ordered cable configurations. Then

$$\mathrm{PathFree}(C,D) = \mathrm{PathFree}(R(C), R(D)). \tag{33}$$

Proof. The transition from C to D connects each point p_n to the corresponding point q_n. If both configurations are reversed, the same geometric correspondence is preserved under the index substitution $n \mapsto N-n+1$. Therefore, the swept region generated by the ordered pair (C,D) is identical to the swept region generated by the ordered pair $(R(C), R(D))$. Since the obstacle set is unchanged, the collision result is also unchanged. Hence

$$\mathrm{PathFree}(C,D) = \mathrm{PathFree}(R(C), R(D)). \tag{34}$$

In contrast, the same invariance does not hold when only one configuration is reversed. In general,

$$\mathrm{PathFree}(C,D) \neq \mathrm{PathFree}(R(C), D), \tag{35}$$

and

$$\mathrm{PathFree}(C,D) \neq \mathrm{PathFree}(C, R(D)). \tag{36}$$

Reversing only one configuration changes the endpoint correspondence during the local deformation. Consequently, the swept envelope may become larger, may become self-intersecting, or may cover a different part of the workspace. Therefore, a transition that is collision-free under one ordering may become invalid under another ordering.

For this reason, transition feasibility must be evaluated for the actual ordered representatives used by the planner. We introduce the endpoint-order variable

$$\sigma \in \{+1, -1\}, \tag{37}$$

where

$$C^{\sigma} = \begin{cases} C, & \sigma = +1, \\ R(C), & \sigma = -1. \end{cases} \tag{38}$$

An ordered transition from C^{σ_C} to D^{σ_D} is admissible only if the swept-envelope condition is satisfied for this particular choice of ordering:

$$\mathrm{PathFree}\,(C^{\sigma_C}, D^{\sigma_D}) = 1. \tag{39}$$

In addition to collision-free deformation, the ordered transition should preserve endpoint consistency. The ordered endpoint axis of a cable configuration is defined as

$$u(C) = \frac{p_N - p_1}{\|p_N - p_1\|}. \tag{40}$$

For two ordered configurations C and D, the change in endpoint direction is

$$\theta(C,D) = \cos^{-1}\left(u(C)^{\top} u(D)\right). \tag{41}$$

The ordered transition is considered endpoint-consistent if

$$\theta(C,D) \le \theta_{\max}, \tag{42}$$

where $\theta_{\max}$ is the maximum admissible change in the ordered cable axis.

For compact notation, we define the ordered edge-feasibility predicate

$$\Gamma(C,D) = 1 \tag{43}$$

if and only if

$$\begin{aligned} &\mathrm{PathFree}(C,D) = 1, \\ &\theta(C,D) \le \theta_{\max}, \\ &\Psi(C,D) = 1 \end{aligned} \tag{44}$$

Here, $\Psi(C,D)$ denotes robot-dependent endpoint feasibility, including admissible endpoint displacement, endpoint obstacle clearance, and end-effector motion limits. If robotic execution constraints are not considered during planning, $\Psi(C,D)$ can be set identically equal to one.

Thus, the ordering used by each accepted tree edge must be fixed during planning, and the ordered transition predicate $\Gamma(\cdot,\cdot)$ must be evaluated before inserting a new edge into the search tree.

4. Global Cable Routing Algorithm

The implemented GCR planner is a bidirectional sampling-based planner in the discretized cable configuration space. It grows one tree from C_{start} and one tree from C_{goal}. Each accepted vertex is a statically valid cable configuration, and each accepted edge satisfies the ordered transition predicate $\Gamma(\cdot,\cdot)$.

4.1. Search Trees and Bidirectional Expansion

Let

$$T_s = (V_s, E_s), \qquad T_g = (V_g, E_g) \tag{45}$$

denote the start and goal trees. They are initialized as

$$V_s = \{C_{\text{start}}\}, \qquad V_g = \{C_{\text{goal}}\}. \tag{46}$$

The algorithm alternates between the two trees. At odd iterations, T_s is expanded toward C_{goal}. At even iterations, T_g is expanded toward C_{start}.

For a new accepted configuration C^{new} with parent C^{par}, the stored cost-to-come is

$$g(C^{\text{new}}) = g(C^{\text{par}}) + d(C^{\text{par}}, C^{\text{new}}) + \alpha_{\text{def}} E_{\text{def}}(C^{\text{new}}, C^{\text{par}}), \quad (47)$$

where $\alpha_{\text{def}} \geq 0$ weights the deformation term. This cost is used as an internal edge-quality measure and is consistent with the path objective $J(\mathscr{P})$ in (**??**).

4.2. Guided Configuration Sampling

At each tree-expansion step, the planner generates a raw candidate cable configuration before applying APF biasing and elastic relaxation. Instead of sampling only from a uniform distribution over the configuration space, the proposed method uses a guided mixture sampler. This sampler combines directed sampling toward the target, local expansion from the active tree, and broad exploration of the workspace.

The raw configuration is sampled from the mixture distribution

$$C^{\text{raw}} \sim p_{\text{goal}}\, \mathscr{Q}_{\text{goal}} + p_{\text{tree}}\, \mathscr{Q}_{\text{tree}} + p_{\text{broad}}\, \mathscr{Q}_{\text{broad}}, \quad (48)$$

where $\mathscr{Q}_{\text{goal}}$, $\mathscr{Q}_{\text{tree}}$, and $\mathscr{Q}_{\text{broad}}$ denote the goal-guided, tree-guided, and broad exploration sampling distributions, respectively. Their probabilities satisfy

$$p_{\text{goal}} + p_{\text{tree}} + p_{\text{broad}} = 1. \quad (49)$$

The goal-guided component $\mathscr{Q}_{\text{goal}}$ generates candidate configurations in the neighborhood of the current target configuration C^{target}. This component increases the probability of producing samples that move the search toward the opposite tree. A small perturbation is added to avoid repeatedly generating identical configurations and to preserve local exploration around the target shape.

The tree-guided component $\mathscr{Q}_{\text{tree}}$ expands from an existing configuration in the active tree. Let $C^p \in V_{\text{act}}$ be a selected parent configuration and let C^ℓ be a local target configuration. The base configuration is generated by moving from C^p toward C^ℓ:

$$C^{\text{base}} = C^p + \alpha \left(C^\ell - C^p \right), \quad (50)$$

where $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ controls the local expansion step. It is chosen according to the distance between C^p and C^ℓ:

$$\alpha = \min \left\{ \alpha_{\max}, \max \left\{ \alpha_{\min}, \frac{\gamma \varepsilon}{d(C^p, C^\ell)} \right\} \right\}, \quad (51)$$

where ε is the local connection radius and $\gamma > 0$ is a scaling coefficient. This rule keeps the sampled configuration within a controlled neighborhood of the active tree while still directing the expansion toward a useful local target.

After a base configuration has been selected, local perturbations are applied to increase sampling diversity. For each cable node, the sampled point is written as

$$p_i^{\text{raw}} = p_i^{\text{base}} + \xi_i + a \sin \left(\frac{\pi (i-1)}{N-1} \right) n_i(C^{\text{base}}), \quad (52)$$

where ξ_i is a small zero-mean Gaussian perturbation, a is a scalar lateral perturbation amplitude, and $n_i(C^{\text{base}})$ is the local normal direction of the cable at node i. The sinusoidal factor makes the lateral deformation smooth along the cable and reduces perturbations near the endpoints.

The broad exploration component $\mathscr{Q}_{\text{broad}}$ generates a new cable shape using a random-walk construction. Starting from an initial node, successive cable nodes are generated as

$$p_{k+1} = p_k + \ell_0 \begin{bmatrix} \cos \theta_k \\ \sin \theta_k \end{bmatrix}, \qquad k = 1, \ldots, N-1, \quad (53)$$

with the heading angle updated according to

$$\theta_{k+1} = \theta_k + \eta_k, \qquad \eta_k \sim \mathscr{N}(0, \sigma_\theta^2). \quad (54)$$

Here, ℓ_0 is the nominal segment length and σ_θ controls the smoothness of the generated shape. After generation, the cable is shifted into the workspace, clipped to the workspace boundaries if necessary, and projected toward the nominal segment length.

The guided mixture sampler therefore balances exploitation and exploration. The goal-guided component improves convergence toward the target, the tree-guided component supports controlled local growth from existing feasible configurations, and the broad component helps the planner escape local geometric restrictions by introducing more diverse cable shapes.

4.3. Configuration-Level Artificial Potential Field Biasing

After a raw cable configuration is generated by the guided sampler, it is biased using a configuration-level artificial potential field (APF). The purpose of this step is to move the sampled cable toward a more promising region of the configuration space and push it away from obstacles before the constrained relaxation stage.

Let

$$C^{\text{raw}} = (p_1^{\text{raw}}, p_2^{\text{raw}}, \ldots, p_N^{\text{raw}})$$

be the raw configuration obtained from the guided mixture sampler introduced in (**??**). It is not inserted directly into the search tree. Instead, it is first transformed into an APF-biased configuration

$$\widehat{C} = (\widehat{p}_1, \widehat{p}_2, \ldots, \widehat{p}_N)$$

Since a cable configuration is represented as

$$C = (p_1, p_2, \ldots, p_N) \in \mathbb{R}^{2N}, \quad (55)$$

the potential field is defined at the configuration level as a scalar function

$$U : \mathbb{R}^{2N} \to \mathbb{R}. \quad (56)$$

Its gradient is

$$\nabla U(C) = \left(\frac{\partial U}{\partial p_1}, \frac{\partial U}{\partial p_2}, \ldots, \frac{\partial U}{\partial p_N} \right), \quad (57)$$

and the corresponding artificial force applied to the n-th cable point is

$$F_n(C) = -\frac{\partial U}{\partial p_n} \quad (58)$$

The total configuration-level potential is defined as

$$U(C) = U_{\text{att}}(C, C_{\text{target}}) + U_{\text{rep}}(C, \mathscr{O}) \quad (59)$$

Here, C_{target} is chosen according to the direction of the bidirectional expansion. When the start tree is expanded, the target configuration is the goal-side configuration. Conversely, when the goal tree is expanded, the target configuration is the start-side configuration. This bidirectional use of the APF is illustrated in

Fig. **??**. Figure **??** shows the APF used for the start-tree expansion, where the attractive component pulls the sampled configuration toward C_{goal}. Figure **??** shows the opposite case, where the goal tree is biased toward C_{start}. In both cases, obstacle regions generate high potential values, forming repulsive zones around occupied areas.

The attractive term guides the cable toward the target configuration:

$$U_{\text{att}}(C, C_{\text{target}}) = \frac{k_{\text{att}}}{2} \sum_{n=1}^{N} \left\| p_n - p_n^{\text{target}} \right\|^2 \tag{60}$$

where $k_{\text{att}} > 0$ is the attractive gain and

$$C_{\text{target}} = \left(p_1^{\text{target}}, p_2^{\text{target}}, \ldots, p_N^{\text{target}} \right) \tag{61}$$

The repulsive term pushes the cable away from obstacles. Since collisions may occur along cable segments and not only at cable nodes, each segment is sampled at $Q+1$ points. For segment $s_k(C)$, the q-th sampled point is

$$s_{kq}(C) = (1-\rho_q)p_k + \rho_q p_{k+1}, \;\; \rho_q = \frac{q}{Q}, \;\; q = 0, \ldots, Q \tag{62}$$

The distance from this point to the obstacle region is

$$r_{kq}(C) = \text{dist}\left(s_{kq}(C), \mathscr{O}\right) = \min_{x \in \mathscr{O}} \left\| s_{kq}(C) - x \right\| \tag{63}$$

The repulsive potential is then written as

$$U_{\text{rep}}(C, \mathscr{O}) = \sum_{k=1}^{N-1} \sum_{q=0}^{Q} \beta \exp\left(-\frac{r_{kq}^2(C)}{2\sigma^2} \right), \tag{64}$$

where $\beta > 0$ controls the repulsion strength and $\sigma > 0$ controls the obstacle influence range.

Starting from C^{raw}, the APF-biased sample is generated by H gradient descent steps:

$$C^{(h+1)} = C^{(h)} - \eta \nabla U(C^{(h)}), \qquad h = 0, \ldots, H-1, \tag{65}$$

with

$$C^{(0)} = C^{\text{raw}}, \qquad \widehat{C} = C^{(H)}. \tag{66}$$

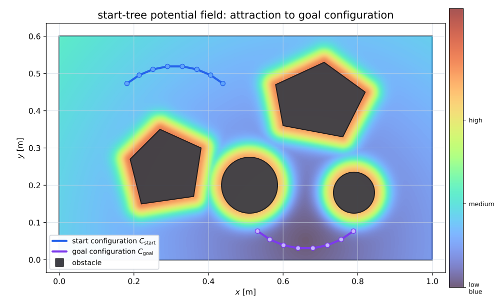
Equivalently, each cable node is updated as

$$p_n^{(h+1)} = p_n^{(h)} + \eta F_n(C^{(h)}), \qquad n = 1, \ldots, N. \tag{67}$$

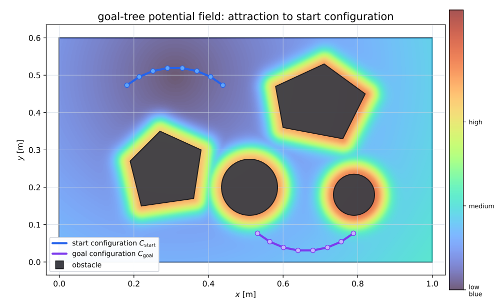
where $\eta > 0$ is the APF step size, which controls the magnitude of the displacement applied to each cable node during one gradient descent iteration. A small value of η produces slow but stable motion in the potential field, whereas a large value may accelerate convergence but can lead to overshooting or unstable updates.

The local effect of this update is shown in Fig. **??**. First, a random configuration C^{raw} is generated in the workspace, as shown in Fig. **??**. Then, the gradient of the total potential produces pointwise artificial forces acting on the cable nodes, as shown in Fig. **??**. These forces combine attraction toward the target configuration with repulsion from obstacles. The raw sample is then transformed into an APF-biased configuration $\widehat{C}$, as shown in Fig. **??**. The biased configuration is closer to the desired expansion direction and is less likely to pass through high-potential obstacle regions.

The resulting configuration $\widehat{C}$ is not guaranteed to satisfy all cable constraints, such as exact segment-length preservation, admissible deformation, or complete collision avoidance. Therefore, it is used as an input for the subsequent constrained relaxation step, where deformation and collision-related violations are reduced before the final validity checks.



(a) Start-tree APF with attraction toward the goal configuration.



(b) Goal-tree APF with attraction toward the start configuration.

Figure 2: Bidirectional configuration-level APF used during tree expansion. The color map represents the normalized potential intensity. Obstacle regions and their neighborhoods correspond to high potential values, while the attractive component biases the expansion toward the opposite tree.

4.4. Cable Elastic Relaxation

The APF-biased configuration $\widehat{C}$ is used as a candidate for tree expansion. However, this candidate may locally violate obstacle-clearance requirements or deformation limits. Therefore, before inserting the candidate into the search tree, a projected elastic relaxation step is applied. The purpose of this step is to obtain a corrected configuration

$$C^{\text{relax}} = (p_1^{\text{relax}}, p_2^{\text{relax}}, \ldots, p_N^{\text{relax}})$$

that remains close to $\widehat{C}$, while preserving the local shape of a nearby feasible reference configuration C^{ref}.

Let $C = (p_1, p_2, \ldots, p_N)$ be the configuration optimized during relaxation. The relaxation objective is defined as

$$\Phi(C) = \lambda_{\text{apf}} \sum_{n=1}^{N} \left\| p_n - \widehat{p}_n \right\|^2 + E_{\text{def}}(C, C^{\text{ref}}) \tag{68}$$

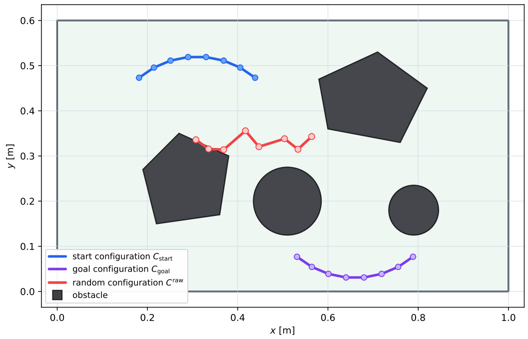
The first term keeps the relaxed configuration close to the APF-biased configuration. The deformation term $E_{\text{def}}(C, C^{\text{ref}})$ contains the segment-vector and discrete-curvature penalties defined in (**??**); these terms reduce excessive stretching, compression, local rotation, and abrupt bending relative to the reference configuration. The weight $\lambda_{\text{apf}} > 0$ controls the relative influence of tracking the APF-biased sample.

The relaxation is initialized from the APF-biased configuration:

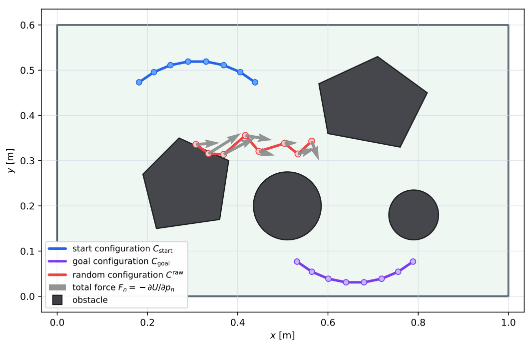
$$C^0 = \widehat{C}. \tag{69}$$

At iteration m, a gradient step is first applied:

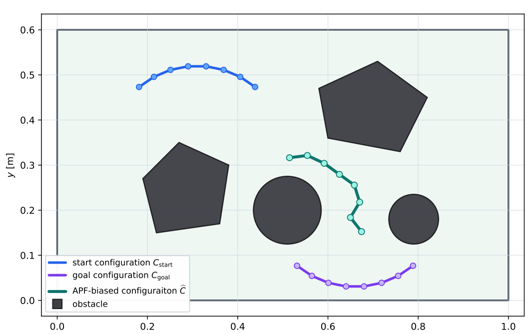
$$C^{m+\frac{1}{2}} = C^m - h_{\text{relax}} \nabla \Phi(C^m), \tag{70}$$



(a) Sampled raw configuration C^{raw}.



(b) Pointwise artificial forces $F_n(C)$ acting on cable nodes.



(c) APF-biased configuration $\widehat{C}$ after gradient descent.

Figure 3: Pipeline of configuration-level APF sampling. A raw random configuration is first generated, then the total APF force is computed for each cable node, and finally the configuration is shifted by gradient descent toward a more promising expansion target.

where $h_{\text{relax}} > 0$ is the relaxation step size. This step reduces the elastic objective and moves the cable toward a configuration that is closer to the APF-biased sample and smoother with respect to the reference shape.

The intermediate configuration is then corrected by a sequence of projection operators:

$$C^{m+1} = \Pi_{\text{end}} \Pi_{\mathscr{W}} \Pi_{\delta} \Pi_{\text{obs}} \left(C^{m+\frac{1}{2}} \right). \tag{71}$$

Each projection has a specific role. The obstacle projection Π_{obs} pushes cable samples away from obstacles when the required clearance is violated. The segment-deviation projection Π_{δ} enforces an upper bound on local deformation with respect to the reference shape. The workspace projection $\Pi_{\mathscr{W}}$ keeps all cable nodes inside the bounded planning workspace. Finally, the endpoint projection Π_{end} restores the APF-updated endpoint positions, so that only the internal cable nodes are modified by the relaxation process.

Obstacle clearance is evaluated at sampled points along each cable segment:

$$x_{kq}(C) = (1-\rho_q) p_k + \rho_q p_{k+1}, \qquad \rho_q = \frac{q}{Q}, \qquad q = 0, \ldots, Q. \tag{72}$$

The clearance condition is

$$d_{\mathscr{O}}\left(x_{kq}(C) \right) \geq d_{\text{safe}}, \tag{73}$$

where $d_{\mathscr{O}}(\cdot)$ is the signed obstacle distance and $d_{\text{safe}} > 0$ is the required safety margin. If a sampled point violates this condition, Π_{obs} applies a local correction in the outward obstacle-normal direction. This correction is distributed to the two points of the corresponding cable segment. Thus, the projection improves clearance without globally reshaping the entire cable.

The segment-deviation projection limits the change of each local segment vector relative to the reference configuration. For each segment, the admissible deviation is

$$\left\| \left(p_{k+1} - p_k \right) - \left(p_{k+1}^{\text{ref}} - p_k^{\text{ref}} \right) \right\| \leq \delta_{\ell}, \qquad k = 1, \ldots, N-1, \tag{74}$$

where $\delta_{\ell} > 0$ is the maximum allowed segment-vector deviation. If this bound is exceeded, Π_{δ} shortens the deviation vector to the radius δ_{ℓ} and updates the corresponding segment nodes accordingly.

The workspace projection enforces the rectangular workspace bounds

$$\mathscr{W} = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}].$$

For a point $p_n = (x_n, y_n)$, this projection clips the coordinates to the workspace limits:

$$\Pi_{\mathscr{W}}(p_n) = \left(\text{clip}(x_n, x_{\min}, x_{\max}), \text{clip}(y_n, y_{\min}, y_{\max}) \right), \tag{75}$$

where

$$\text{clip}(a, a_{\min}, a_{\max}) = \min\left\{ a_{\max}, \max\{a_{\min}, a\} \right\} \tag{76}$$

The endpoint projection Π_{end} fixes the two cable endpoints:

$$p_1 = \widehat{p}_1, \qquad p_N = \widehat{p}_N \tag{77}$$

This preserves the endpoint displacement generated by the APF step while allowing the internal cable nodes to relax.

After M_{relax} iterations, the relaxed configuration is defined as

$$C^{\text{relax}} = C^{M_{\text{relax}}}. \tag{78}$$

The resulting candidate C^{relax} is considered for insertion into the search tree only if it passes the final admissibility check,

$$C^{\text{relax}} \in \mathscr{C}_{\text{free}} \tag{79}$$

This check verifies workspace bounds, segment-deviation limits, and collision freedom of all cable segments.

The effect of the relaxation step is illustrated in Fig. **??**. The APF-biased configuration is locally corrected so that the final candidate remains close to the APF output while satisfying the geometric and deformation constraints required for subsequent transition validation.

4.5. Feasible Parent Selection and Node Insertion

After the relaxed configuration C^{relax} has passed the static admissibility check in **??**, the planner searches for a feasible parent in

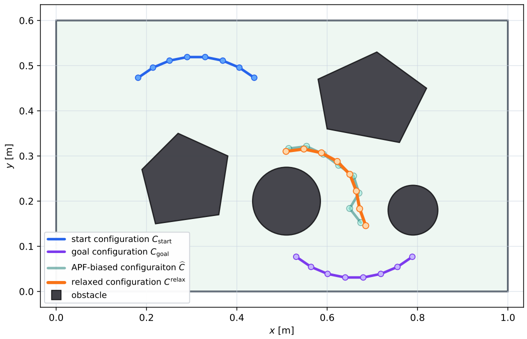


Figure 4: Projected elastic relaxation of an APF-biased cable configuration. The relaxed configuration remains close to the APF-biased sample while improving obstacle clearance and satisfying local segment-deviation constraints.

the active tree. The parent selection step determines whether the relaxed configuration can be connected to an existing tree vertex by a collision-free and endpoint-consistent local deformation.

In contrast to static configuration validation, local transition validation depends on the ordering of the cable nodes. Therefore, the relaxed configuration is not inserted into the tree only as an unordered geometric curve. Instead, the planner considers the admissible ordered representatives of the relaxed configuration. Using the reversal operator $R(\cdot)$ introduced in Section **??**, the set of possible ordered representatives is

$$\mathscr{R}\left(C^{\mathrm{relax}}\right) = \left\{C^{\mathrm{relax}}, R\left(C^{\mathrm{relax}}\right)\right\}. \tag{80}$$

Equivalently, for $\sigma \in \{+1, -1\}$, we write

$$\left(C^{\mathrm{relax}}\right)^{\sigma} = \begin{cases} C^{\mathrm{relax}}, & \sigma = +1, \\ R\left(C^{\mathrm{relax}}\right), & \sigma = -1. \end{cases} \tag{81}$$

Each vertex already stored in the active tree has a fixed endpoint ordering, because it was inserted through a previously validated ordered transition.

Let V_{act} denote the set of vertices in the active tree. A pair (C_i, σ), where $C_i \in V_{\mathrm{act}}$ and $\sigma \in \{+1, -1\}$, is first considered as a parent-candidate pair only if the corresponding ordered relaxed configuration lies within the local connection radius:

$$d\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right) \leq \varepsilon. \tag{82}$$

This condition restricts the parent search to nearby configurations and avoids large local deformations that may be difficult to validate. The set of nearby parent-candidate pairs is therefore

$$\mathscr{V}_{\varepsilon} = \left\{(C_i, \sigma) \,\middle|\, C_i \in V_{\mathrm{act}},\ \sigma \in \{+1, -1\},\ d\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right) \leq \varepsilon\right\}. \tag{83}$$

The candidates in $\mathscr{V}_{\varepsilon}$ are ordered according to their distance from the corresponding ordered relaxed configuration. Only a bounded number of nearest candidates is evaluated. This keeps the parent-selection step local and computationally bounded.

Distance proximity alone is not sufficient for parent selection. A candidate parent must also produce a deformation-feasible edge under the selected endpoint ordering. For each candidate pair (C_i, σ), let

$$\left(C^{\mathrm{relax}}\right)^{\sigma} = \left(p_1^{\mathrm{relax},\sigma}, p_2^{\mathrm{relax},\sigma}, \ldots, p_N^{\mathrm{relax},\sigma}\right).$$

The swept envelope between C_i and $\left(C^{\mathrm{relax}}\right)^{\sigma}$ is constructed as

$$\mathscr{E}_{i,\mathrm{relax}}^{\sigma} = \mathrm{Poly}\left(p_1^i, p_2^i, \ldots, p_N^i, p_N^{\mathrm{relax},\sigma}, p_{N-1}^{\mathrm{relax},\sigma}, \ldots, p_1^{\mathrm{relax},\sigma}\right). \tag{84}$$

The candidate parent is feasible only if this swept envelope does not intersect the obstacle region:

$$\mathscr{E}_{i,\mathrm{relax}}^{\sigma} \cap \mathscr{O} = \emptyset. \tag{85}$$

Equivalently, the transition-validity condition is written as

$$\mathrm{PathFree}\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right) = 1. \tag{86}$$

Since reversing only one configuration can change the point-to-point correspondence during the transition, the conditions

$$\mathrm{PathFree}\left(C_i, C^{\mathrm{relax}}\right) = 1$$

and

$$\mathrm{PathFree}\left(C_i, R\left(C^{\mathrm{relax}}\right)\right) = 1$$

are not generally equivalent. Therefore, the parent selection step evaluates the swept-envelope condition for the actual ordered representative that would be inserted into the tree.

To include ordered edge-feasibility consistency in the same edge-validation step, the ordered transition predicate $\Gamma(\cdot,\cdot)$ introduced in Section **??** is used. Thus, the feasible parent set is defined as

$$\mathscr{V}_{\mathrm{par}} = \left\{(C_i, \sigma) \in \mathscr{V}_{\varepsilon} \,\middle|\, \Gamma\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right) = 1\right\}. \tag{87}$$

Here, the predicate Γ includes the ordered swept-envelope condition, the endpoint-axis continuity condition, and, when required, the robot-dependent endpoint feasibility condition. In particular,

$$\Gamma\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right) = 1 \implies \mathrm{PathFree}\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right) = 1. \tag{88}$$

If $\mathscr{V}_{\mathrm{par}} = \emptyset$, the relaxed configuration is rejected because no valid ordered local deformation connects it to the active tree.

If at least one feasible parent exists, the parent and the ordering of the relaxed configuration are selected from $\mathscr{V}_{\mathrm{par}}$. In the simplest implementation, the nearest feasible ordered parent-candidate pair is selected:

$$(C^{\mathrm{par}}, \sigma^*) = \arg \min_{(C_i,\sigma) \in \mathscr{V}_{\mathrm{par}}} d\left(C_i, \left(C^{\mathrm{relax}}\right)^{\sigma}\right). \tag{89}$$

The relaxed configuration is then inserted into the active tree using the selected endpoint ordering:

$$C^{\mathrm{new}} = \left(C^{\mathrm{relax}}\right)^{\sigma^*}. \tag{90}$$

The corresponding edge

$$(C^{\mathrm{par}}, C^{\mathrm{new}}) \tag{91}$$

is added to the active tree, and the parent pointer of C^{new} is set to C^{par}.

By construction, every inserted edge satisfies

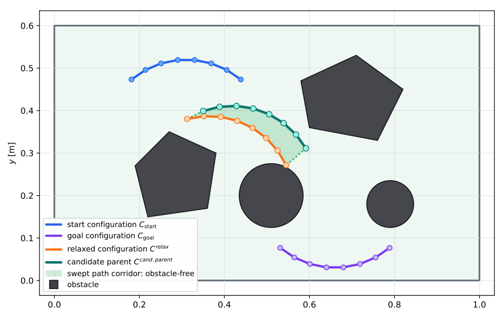
$$\Gamma\left(C^{\mathrm{par}}, C^{\mathrm{new}}\right) = 1, \tag{92}$$

and therefore also satisfies

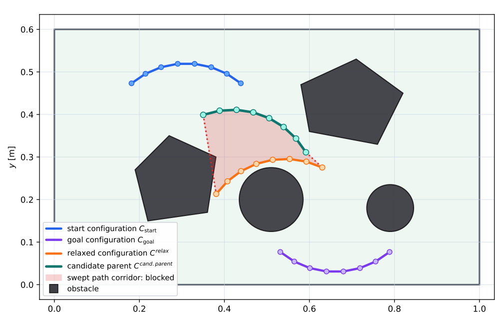
$$\mathrm{PathFree}\left(C^{\mathrm{par}}, C^{\mathrm{new}}\right) = 1. \tag{93}$$

Thus, each accepted edge represents an obstacle-free swept deformation under the selected endpoint correspondence. This prevents a configuration from being accepted under one endpoint ordering and later reinterpreted using another ordering that was not validated during planning.

The two possible outcomes of the parent-selection test are illustrated in Fig. **??**. In Fig. **??**, the swept corridor between the candidate parent and the ordered relaxed configuration is obstacle-free. Therefore, the candidate parent is accepted and the ordered relaxed configuration can be inserted into the active tree. In Fig. **??**, the swept corridor intersects an obstacle. Although the two endpoint configurations may be individually valid, the ordered deformation between them is not collision-free, and the candidate parent is rejected.



(a) Feasible parent connection. The swept corridor between the candidate parent $C^{\text{cand.parent}}$ and the ordered relaxed configuration C^{new} is obstacle-free.



(b) Rejected parent connection. The swept corridor intersects an obstacle, so the transition from the candidate parent to the ordered relaxed configuration is not path-free.

Figure 5: Feasible parent selection using the swept envelope between a candidate parent configuration and the ordered relaxed configuration. The relaxed configuration can be inserted into the active tree only if at least one nearby parent produces an obstacle-free and endpoint-consistent swept corridor.

4.6. Tree Connection and Path Extraction

After a new node C^{new} has been inserted into the active tree, the algorithm attempts to connect it to the opposite tree. Since each tree vertex is stored with a fixed endpoint ordering, the connection step must evaluate the ordered transition between the two meeting configurations. Therefore, the connection between the two trees is validated using the same ordered edge-feasibility predicate $\Gamma(\cdot,\cdot)$ used in the parent-selection stage.

Let V_{opp} denote the vertex set of the opposite tree. The nearest candidate connection node is first defined as

$$C_{\text{conn}} = \arg\min_{C \in V_{\text{opp}}} d(C, C^{\text{new}}). \tag{94}$$

This nearest-neighbor query provides a candidate connection between the two trees. However, distance proximity alone is not sufficient, because the transition from one tree to the other must preserve both swept-collision validity and ordered edge-feasibility consistency.

If the active tree is T_{s}, then the final path direction across the connection edge is from C^{new} to C_{conn}. Therefore, a valid connection must satisfy

$$d(C^{\text{new}}, C_{\text{conn}}) \le \varepsilon, \tag{95}$$

and

$$\Gamma(C^{\text{new}}, C_{\text{conn}}) = 1. \tag{96}$$

Equivalently, this implies

$$\text{PathFree}(C^{\text{new}}, C_{\text{conn}}) = 1. \tag{97}$$

If the active tree is T_{g}, then C^{new} belongs to the goal-side tree, while C_{conn} belongs to the start-side tree. In the final extracted path, the connection edge is traversed from C_{conn} to C^{new}. Therefore, the ordered connection condition is written as

$$d(C_{\text{conn}}, C^{\text{new}}) \le \varepsilon, \tag{98}$$

and

$$\Gamma(C_{\text{conn}}, C^{\text{new}}) = 1. \tag{99}$$

The use of $\Gamma(\cdot,\cdot)$ is necessary because the two meeting configurations are ordered configurations, not only geometric cable configurations. Although two configurations may be individually collision-free, an invalid endpoint correspondence may generate a different swept envelope and may therefore create a collision during the connection transition. Thus, the connection edge is accepted only if the actual ordered transition used in the final path is feasible. For implementation, if the nearest candidate C_{conn} does not satisfy the ordered connection condition, the planner may evaluate a bounded set of nearby vertices from V_{opp}. This set is defined as

$$\mathscr{V}_{\text{conn}} = \left\{ C_j \in V_{\text{opp}} \;\middle|\; d(C_j, C^{\text{new}}) \le \varepsilon \right\}. \tag{100}$$

When the active tree is T_{s}, the feasible connection set is

$$\mathscr{V}_{\text{conn}}^{\text{feas}} = \left\{ C_j \in \mathscr{V}_{\text{conn}} \;\middle|\; \Gamma(C^{\text{new}}, C_j) = 1 \right\}. \tag{101}$$

When the active tree is T_{g}, the feasible connection set is

$$\mathscr{V}_{\text{conn}}^{\text{feas}} = \left\{ C_j \in \mathscr{V}_{\text{conn}} \;\middle|\; \Gamma(C_j, C^{\text{new}}) = 1 \right\}. \tag{102}$$

If

$$\mathscr{V}_{\text{conn}}^{\text{feas}} = \emptyset, \tag{103}$$

then no valid tree connection is made during the current iteration. Otherwise, the connection node is selected as the nearest feasible connection candidate:

$$C_{\text{conn}} = \arg\min_{C_j \in \mathscr{V}_{\text{conn}}^{\text{feas}}} d(C_j, C^{\text{new}}). \tag{104}$$

If the active tree is T_{s}, the meeting nodes are assigned as

$$C_{\text{meet}}^{\text{s}} = C^{\text{new}}, \qquad C_{\text{meet}}^{\text{g}} = C_{\text{conn}}. \tag{105}$$

If the active tree is T_{g}, the assignment is

$$C_{\text{meet}}^{\text{s}} = C_{\text{conn}}, \qquad C_{\text{meet}}^{\text{g}} = C^{\text{new}}. \tag{106}$$

The start-side partial path is obtained by tracing parent pointers from $C_{\text{meet}}^{\text{s}}$ to the root C_{start} and reversing the order:

$$\mathscr{P}_{\text{s}} = \text{Reverse}\left(\text{TraceParents}\left(C_{\text{meet}}^{\text{s}} \to C_{\text{start}}\right)\right). \tag{107}$$

The goal-side partial path is obtained by tracing parent pointers from C^{g}_{meet} to the root C_{goal}:

$$\mathscr{P}_{\mathrm{g}} = \mathrm{TraceParents}\left(C^{g}_{\mathrm{meet}} \to C_{\mathrm{goal}}\right). \tag{108}$$

Thus, the complete solution path is

$$\mathscr{P} = \mathscr{P}_{\mathrm{s}} \oplus \mathscr{P}_{\mathrm{g}}, \tag{109}$$

where $\oplus$ denotes path concatenation.

By construction, every configuration in $\mathscr{P}$ belongs to $\mathscr{C}_{\mathrm{free}}$, and every edge between consecutive configurations satisfies the ordered deformation-feasibility condition:

$$\Gamma(C_k, C_{k+1}) = 1, \qquad k = 0, \ldots, |\mathscr{P}| - 2. \tag{110}$$

Consequently,

$$\mathrm{PathFree}(C_k, C_{k+1}) = 1, \qquad k = 0, \ldots, |\mathscr{P}| - 2. \tag{111}$$

Therefore, the extracted solution path is not only a sequence of statically admissible cable configurations, but also a sequence of ordered, endpoint- consistent, swept-collision-free cable transitions.

4.7. Rewiring: Post-Connection Shortcut Refinement

The first path obtained by connecting the two trees may contain unnecessary intermediate configurations. Therefore, after a feasible solution has been found, a shortcut-based refinement step is applied. This step reduces the number of cable configurations while preserving collision and deformation feasibility.

Let the initial connected path be

$$\mathscr{P}^0 = \{C_0, C_1, \ldots, C_K\}, \qquad C_0 = C_{\mathrm{start}}, \qquad C_K = C_{\mathrm{goal}} \tag{112}$$

For any pair of configurations C_i and C_j with $j > i+1$, the algorithm attempts to replace the subsequence

$$(C_i, C_{i+1}, \ldots, C_j) \tag{113}$$

by the direct transition (C_i, C_j). This replacement is accepted if it does not increase the accumulated edge cost,

$$d(C_i, C_j) + \alpha_{\mathrm{def}} E_{\mathrm{def}}(C_j, C_i) \le \sum_{k=i}^{j-1} \left[d(C_k, C_{k+1}) + \alpha_{\mathrm{def}} E_{\mathrm{def}}(C_{k+1}, C_k) \right], \tag{114}$$

and if the direct transition is deformation-feasible:

$$\Gamma(C_i, C_j) = 1 \tag{115}$$

The shortcutting procedure is repeated until no further improvement can be made, producing the compact path

$$\mathscr{P}^{\mathrm{ref}} = \{C_0, C_1, \ldots, C_M\}, \qquad M \le K \tag{116}$$

Since every accepted shortcut must satisfy (**??**), the refined path preserves the collision and transition-feasibility properties enforced by the original path checks.

4.8. Summary of the GCR Planner

The proposed GCR planner is a validity-preserving bidirectional search in the discretized cable configuration space. It grows one tree from the start configuration and one tree from the goal configuration. At each iteration, one tree is selected as the active tree and expanded toward the opposite side. A raw cable configuration is generated by the guided sampler, biased by the configuration-level APF, and corrected by projected elastic relaxation. The relaxed candidate is inserted only if it satisfies static admissibility and if there exists a feasible ordered parent connection.

The main difference between GCR and a standard bidirectional RRT is that node insertion is not based only on distance and static collision checking. In GCR, each accepted vertex must belong to $\mathscr{C}_{\mathrm{free}}$, and each accepted edge must satisfy the ordered edge-feasibility predicate $\Gamma(\cdot,\cdot)$. This predicate includes swept-envelope collision checking, endpoint-axis consistency, and optional robot-dependent endpoint feasibility. Therefore, the extracted path is a sequence of collision-free cable shapes connected by ordered, deformation-feasible transitions.

Algorithm **??** gives the global bidirectional structure of the planner. Algorithm **??** describes the expansion step, including sampling, APF biasing, relaxation, static validation, ordered parent selection, and node insertion. Algorithm **??** summarizes shortcut refinement, where redundant intermediate configurations are removed only when the replacement edge remains ordered-feasible and does not increase the edge cost.

The planner terminates successfully when it finds a finite path

$$\mathscr{P} = \{C_0, C_1, \ldots, C_K\} \tag{117}$$

such that

$$C_0 = C_{\mathrm{start}}, \qquad C_K = C_{\mathrm{goal}}, \tag{118}$$

$$C_k \in \mathscr{C}_{\mathrm{free}}, \qquad k = 0, \ldots, K, \tag{119}$$

and

$$\Gamma(C_k, C_{k+1}) = 1, \qquad k = 0, \ldots, K-1. \tag{120}$$

If no valid connection between the two trees is found within the maximum number of iterations $I_{\max}$, the planner reports failure.

The procedure NEAR returns all vertices of the opposite tree located within the connection radius ε from C^{new}. The procedure FEASIBLECONNECTIONS evaluates the ordered edge predicate $\Gamma(\cdot,\cdot)$. If $\mathrm{dir} = 1$, it accepts vertices satisfying $\Gamma(C^{\mathrm{new}}, C_j) = 1$. If $\mathrm{dir} = -1$, it accepts vertices satisfying $\Gamma(C_j, C^{\mathrm{new}}) = 1$. This distinction preserves the direction of the transition in the final extracted path.

5. Simulation and Validation

We validate the proposed Global Cable Routing (GCR) planner in two stages. First, we evaluate the planner in planar benchmark scenarios with static obstacles. This stage measures success rate, computation time, number of search iterations, and compactness of the resulting configuration sequence. Second, we use a MuJoCo dual-arm manipulation setup in which the robot grippers are rigidly attached to the cable endpoints. This validation checks whether endpoint trajectories generated from the planned cable configurations can reproduce the target cable shape in simulation.

5.1. Benchmark Evaluation

We use the benchmark study to evaluate GCR at the cable-configuration level. Each benchmark scenario contains a bounded planar workspace, a set of static obstacles, an initial cable configuration C_{start}, and a target cable configuration C_{goal}. For each scenario, we run 1000 independent trials with different random seeds. We count a trial as successful when the planner finds a sequence of cable configurations from C_{start} to C_{goal} before reaching the maximum number of iterations. The accepted sequence must satisfy

Algorithm 1 Global Cable Routing Planner

Require: Start configuration C_{start}, goal configuration C_{goal}, obstacle region $\mathscr{O}$, maximum number of iterations $I_{\max}$, connection radius ε
Ensure: Deformation-feasible path $\mathscr{P}$ or failure
Begin:
if $C_{\text{start}} \notin \mathscr{C}_{\text{free}} \vee C_{\text{goal}} \notin \mathscr{C}_{\text{free}}$ **then**
  return failure
end if
$V_s \leftarrow \{C_{\text{start}}\}$, $E_s \leftarrow \emptyset$
$V_g \leftarrow \{C_{\text{goal}}\}$, $E_g \leftarrow \emptyset$
$T_s \leftarrow (V_s, E_s)$, $T_g \leftarrow (V_g, E_g)$
$g(C_{\text{start}}) \leftarrow 0$, $g(C_{\text{goal}}) \leftarrow 0$
for $i = 1, \dots, I_{\max}$ **do**
  if odd(i) **then**
    $T_{\text{act}} \leftarrow T_s$
    $T_{\text{opp}} \leftarrow T_g$
    $C^{\text{target}} \leftarrow C_{\text{goal}}$
    dir $\leftarrow 1$
  else
    $T_{\text{act}} \leftarrow T_g$
    $T_{\text{opp}} \leftarrow T_s$
    $C^{\text{target}} \leftarrow C_{\text{start}}$
    dir $\leftarrow -1$
  end if
  $(\text{success}, C^{\text{new}}) \leftarrow \textsc{ExpandTree}(T_{\text{act}}, C^{\text{target}}, \mathscr{O}, \varepsilon)$
  if success = false **then**
    continue
  end if
  $V_{\text{conn}} \leftarrow \textsc{Near}(T_{\text{opp}}, C^{\text{new}}, \varepsilon)$
  $V_{\text{conn}}^{\text{feas}} \leftarrow \textsc{FeasibleConnections}(V_{\text{conn}}, C^{\text{new}}, \text{dir})$
  if $V_{\text{conn}}^{\text{feas}} = \emptyset$ **then**
    continue
  end if
  $C^{\text{conn}} \leftarrow \textsc{Nearest}(V_{\text{conn}}^{\text{feas}}, C^{\text{new}})$
  if dir $= 1$ **then**
    $C_{\text{meet}}^{s} \leftarrow C^{\text{new}}$
    $C_{\text{meet}}^{g} \leftarrow C^{\text{conn}}$
  else
    $C_{\text{meet}}^{s} \leftarrow C^{\text{conn}}$
    $C_{\text{meet}}^{g} \leftarrow C^{\text{new}}$
  end if
  $\mathscr{P}_s \leftarrow \textsc{TraceParents}(C_{\text{meet}}^{s}, C_{\text{start}})$
  $\mathscr{P}_s \leftarrow \textsc{Reverse}(\mathscr{P}_s)$
  $\mathscr{P}_g \leftarrow \textsc{TraceParents}(C_{\text{meet}}^{g}, C_{\text{goal}})$
  $\mathscr{P} \leftarrow \mathscr{P}_s \oplus \mathscr{P}_g$
  $\mathscr{P} \leftarrow \textsc{ShortcutRefine}(\mathscr{P}, \mathscr{O})$
  return $\mathscr{P}$
end for
return failure

Algorithm 2 Tree Expansion

Require: Active tree $T_{\text{act}} = (V_{\text{act}}, E_{\text{act}})$, target configuration C^{target}, obstacle region $\mathscr{O}$, connection radius ε
Ensure: Pair $(\text{success}, C^{\text{new}})$
Begin:
Sample a raw configuration C^{raw} using the guided mixture sampler
$\widehat{C} \leftarrow \text{APF}(C^{\text{raw}}, C^{\text{target}}, \mathscr{O})$
$C^{\text{ref}} \leftarrow \arg\min_{C_i \in V_{\text{act}}} d(C_i, \widehat{C})$
$C^{\text{relax}} \leftarrow \textsc{Relax}(\widehat{C}, C^{\text{ref}}, \mathscr{O})$
if $C^{\text{relax}} = \emptyset$ **then**
  return $(\text{false}, \emptyset)$
end if
if $C^{\text{relax}} \notin \mathscr{C}_{\text{free}}$ **then**
  return $(\text{false}, \emptyset)$
end if
$V_\varepsilon \leftarrow \left\{(C_i, \sigma) \,\middle|\, C_i \in V_{\text{act}},\ \sigma \in \{+1, -1\},\ d\left(C_i, \left(C^{\text{relax}}\right)^{\sigma}\right) \le \varepsilon\right\}$
$V_{\text{par}} \leftarrow \left\{(C_i, \sigma) \in V_\varepsilon \,\middle|\, \Gamma\left(C_i, \left(C^{\text{relax}}\right)^{\sigma}\right) = 1\right\}$
if $V_{\text{par}} = \emptyset$ **then**
  return $(\text{false}, \emptyset)$
end if
$(C^{\text{par}}, \sigma^*) \leftarrow \arg\min_{(C_i,\sigma) \in V_{\text{par}}} \left[g(C_i) + d\left(C_i, \left(C^{\text{relax}}\right)^{\sigma}\right) + \alpha_{\text{def}} E_{\text{def}}\left(\left(C^{\text{relax}}\right)^{\sigma}, C_i\right)\right]$
$C^{\text{new}} \leftarrow \left(C^{\text{relax}}\right)^{\sigma^*}$
$V_{\text{act}} \leftarrow V_{\text{act}} \cup \{C^{\text{new}}\}$
$E_{\text{act}} \leftarrow E_{\text{act}} \cup \{(C^{\text{par}}, C^{\text{new}})\}$
parent$(C^{\text{new}}) \leftarrow C^{\text{par}}$
$g(C^{\text{new}}) \leftarrow g(C^{\text{par}}) + d(C^{\text{par}}, C^{\text{new}}) + \alpha_{\text{def}} E_{\text{def}}(C^{\text{new}}, C^{\text{par}})$
return $(\text{true}, C^{\text{new}})$

Algorithm 3 Shortcut Refinement

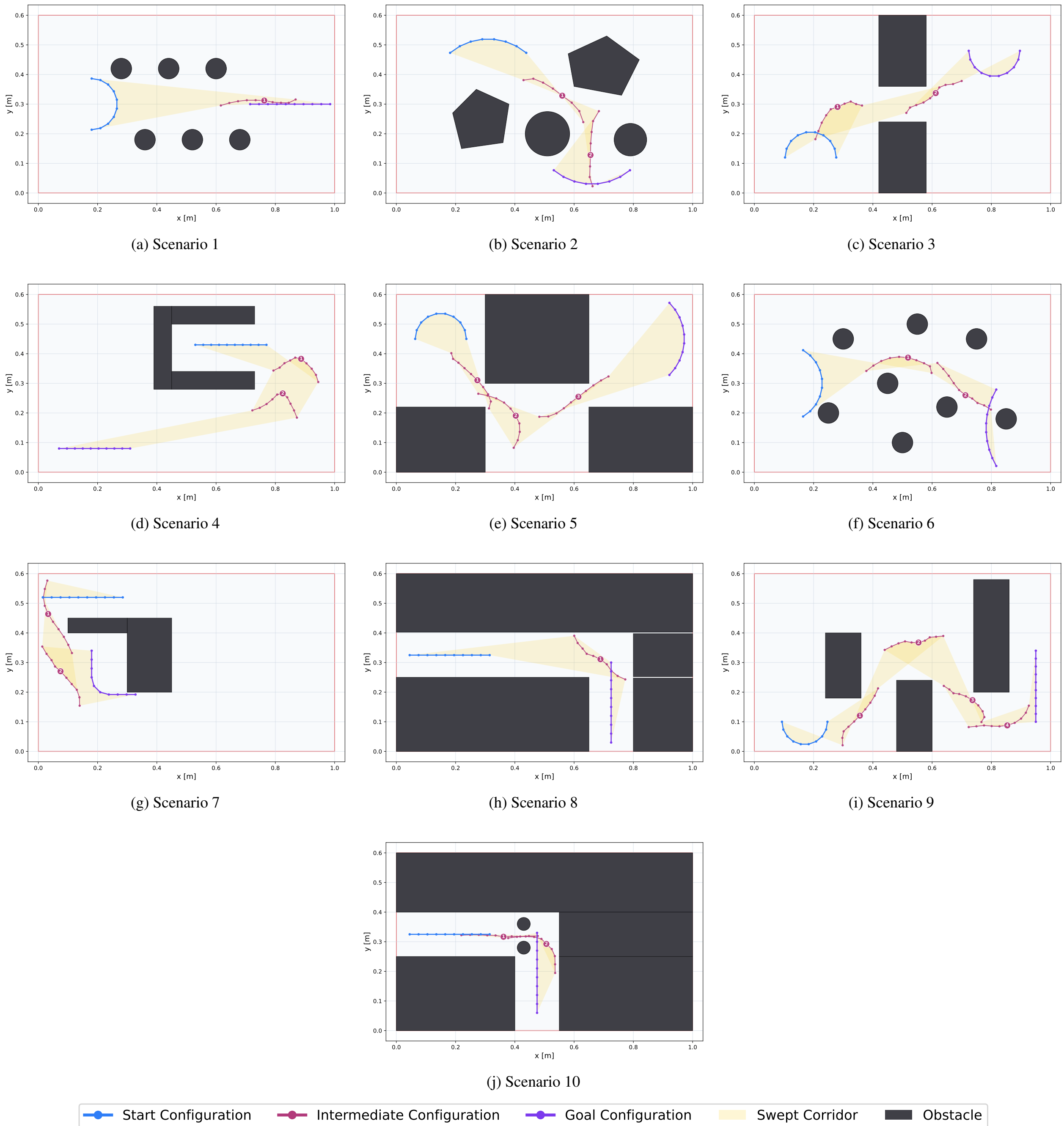
Require: Initial path $\mathscr{P} = \{C_0, C_1, \dots, C_K\}$, obstacle region $\mathscr{O}$
Ensure: Refined path $\mathscr{P}^{\text{ref}}$
Begin:
$\mathscr{P}^{\text{ref}} \leftarrow \mathscr{P}$
$i \leftarrow 0$
while $i < |\mathscr{P}^{\text{ref}}| - 2$ **do**
  $j \leftarrow |\mathscr{P}^{\text{ref}}| - 1$
  while $j > i + 1$ **do**
    $C_i \leftarrow \mathscr{P}^{\text{ref}}[i]$
    $C_j \leftarrow \mathscr{P}^{\text{ref}}[j]$
    $J_{\text{old}} \leftarrow \sum_{k=i}^{j-1} \left[d(C_k, C_{k+1}) + \alpha_{\text{def}} E_{\text{def}}(C_{k+1}, C_k)\right]$
    $J_{\text{new}} \leftarrow d(C_i, C_j) + \alpha_{\text{def}} E_{\text{def}}(C_j, C_i)$
    if $\Gamma(C_i, C_j) = 1$ **and** $J_{\text{new}} \le J_{\text{old}}$ **then**
      Remove configurations $C_{i+1}, \dots, C_{j-1}$ from $\mathscr{P}^{\text{ref}}$
      break
    end if
    $j \leftarrow j - 1$
  end while
  $i \leftarrow i + 1$
end while
return $\mathscr{P}^{\text{ref}}$

the static configuration validity test, the swept-transition feasibility test, and the ordered edge-feasibility validation described in Section **??**.

All benchmark experiments are executed using the single-threaded C++ implementation of GCR. The tests are performed on a laptop equipped with an AMD Ryzen 5 5600H CPU with a 3.30GHz base clock and 24GB of DDR4 memory at 3200MT/s. The reported computation time includes tree expansion, shortcut refinement, and ordered edge-feasibility validation. No GPU acceleration or parallel execution is used in the reported benchmark results.

The benchmark contains ten representative routing scenarios. The scenarios differ in obstacle arrangement, free-space connectivity, passage width, and the amount of deformation required to move the cable from C_{start} to C_{goal}. Some scenarios provide a relatively direct route between the two configurations. Others require the cable to pass through narrow passages, rotate around obstacle boundaries, or follow a less direct deformation sequence. Figure **??** shows one representative solution for each scenario. These visual results are important for interpreting the numerical results because the difficulty of a routing problem depends not only on the number of obstacles, but also on how the obstacles restrict feasible cable transitions.

We report the following metrics for each benchmark scenario:



(a) Scenario 1
(b) Scenario 2
(c) Scenario 3
(d) Scenario 4
(e) Scenario 5
(f) Scenario 6
(g) Scenario 7
(h) Scenario 8
(i) Scenario 9
(j) Scenario 10

Figure 6: Representative GCR solutions for the ten benchmark scenarios. Each subfigure shows the initial cable configuration, the target cable configuration, the static obstacles, and the resulting deformation sequence.

- **Success rate**: the percentage of trials in which GCR finds an accepted path before reaching the maximum number of iterations.
- **Average planning time**: the mean runtime of the complete planning pipeline over successful trials. This includes tree expansion, shortcut refinement, and ordered edge-feasibility validation.
- **Fastest planning time**: the minimum runtime observed among the successful trials. This value indicates the best-case behavior of the stochastic search for a given scenario.
- **Average number of iterations**: the mean number of tree-expansion iterations required to obtain a valid connection between the two trees.
- **Configurations before rewiring**: the average number of cable configurations in the raw path immediately after the two trees are connected.
- **Configurations after rewiring**: the average number of cable configurations remaining after shortcut refinement. This metric describes the compactness of the final path that is later used for execution.

5.2. Benchmark Results

Table **??** summarizes the benchmark results. All ten selected scenarios were solved in all 1000 trials. This result should be interpreted within the tested scenario set: it shows that the implemented combination of bidirectional expansion, APF-biased sampling, elastic relaxation, swept-transition validation, and ordered edge-feasibility validation was sufficient for these planar layouts.

The results in Table **??** show that GCR found a valid path in all tested trials for the ten benchmark scenarios. Each scenario was evaluated over 1000 independent runs, and the success rate remained 100% in every case. This indicates that, for the selected planar environments, the combination of bidirectional search, APF-guided sampling, elastic relaxation, and swept-transition validation consistently produced feasible cable-routing paths.

The computational cost varies with the geometric constraints of the workspace. **Scenario 3**, Figure **??**, is solved with the lowest average computation time, 0.008 ± 0.003 s, and requires only $(0.07 \pm 0.03) \times 10^3$ iterations on average. This scenario represents a relatively simple routing case, where the free space allows a direct deformation sequence between C_{start} and C_{goal}. **Scenario 5**, Figure **??**, also remains computationally light, with an average time of 0.017 ± 0.005 s and $(0.09 \pm 0.03) \times 10^3$ iterations. Most trials are solved quickly, as indicated by the fastest time of 0.005 s.

The most demanding case is **Scenario 4**, Figure **??**, where the planner requires 1.306 ± 0.871 s and $(6.98 \pm 4.52) \times 10^3$ iterations on average. This increase is consistent with the more constrained obstacle arrangement, where the cable has fewer admissible deformation routes and the planner must reject more samples because of static collisions or invalid swept transitions. **Scenario 10**, Figure **??**, is the second most expensive case, with 0.339 ± 0.139 s and $(2.60 \pm 1.08) \times 10^3$ iterations. Despite this higher complexity, both scenarios still maintain a 100% success rate across all trials.

For the remaining scenarios, **Scenarios 1, 2, 6, 7, 8, and 9** shown in Figure **??**, the average runtime ranges from 0.072 s to 0.283 s. These values show that the planner handled the selected static-obstacle tasks efficiently, except when the geometry imposed narrow or highly constrained deformation routes.

The shortcut refinement step also affects the final path representation. Before refinement, the raw connected paths contain between 13 ± 2 and 25 ± 4 configurations on average. After refinement, the final paths contain between 3 and 9 ± 1 configurations. The retained configurations are those that preserve the validity of the cable transition according to the swept-envelope test and ordered edge-feasibility validation.

Overall, the benchmark results show that GCR solves all selected planar routing scenarios with full success over 1000 trials per scenario. The average computation time is below 0.34 s in nine out of ten scenarios, while the most constrained scenario is solved in about 1.31 s on average. These results support the use of GCR as an efficient offline planner for generating compact, collision-free, and transition-feasible cable configuration sequences before robotic execution.

5.3. Robotic Manipulation Validation

The second validation stage evaluates whether the refined GCR paths can guide a simulated dual-arm robotic system to deform and translate a cable. The robotic setup, built in MuJoCo **?**, is shown in Fig. **??**. It contains two UR10 manipulators attached rigidly to the two cable ends, the target cable shape, and the bounded planar workspace with obstacles. The robots track the planned endpoint trajectories, while the internal cable elements move passively according to the simulated cable dynamics, elasticity, gravity, contact response, and endpoint motion.

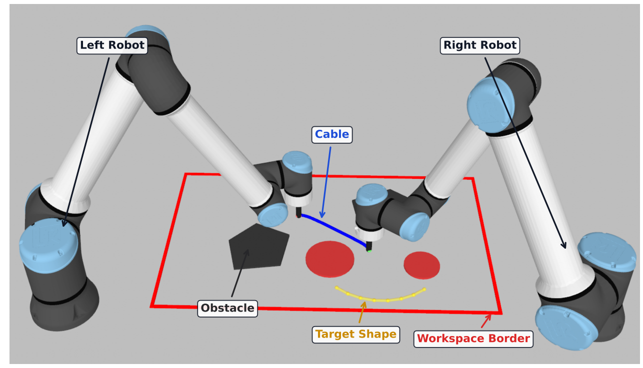


Figure 7: Dual-arm MuJoCo setup used for simulation-based manipulation validation.

For each waypoint configuration, the first and last cable points define the Cartesian references for the left and right robot end-effectors. The endpoint motion between two consecutive configurations is executed using smooth task-space interpolation. The settled cable configuration at the end of the execution is then compared with the planned goal configuration.

5.4. Robotic Execution Results

Table **??** reports the execution accuracy for the scenarios. The root mean square error (RMSE) and maximum point-wise error ($e_{\max}$) are computed over all cable nodes between the settled simulation state and the goal configuration.

$$e_{\max} = \max(e_1, e_2, \ldots, e_N) \tag{121}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} e_n^2} \tag{122}$$

where e_n is the Euclidean error between the n-th point of the final settled configuration and the corresponding point of the target configuration.

The simulated executions produce final RMSE configuration errors between 1.60 mm and 4.66 mm. For the considered cable length $L = 0.27$ m, this corresponds to approximately 0.59–1.73% of the cable length. The maximum point-wise error ranges from 2.31 mm to 7.81 mm, or approximately 0.86–2.89% of the cable length. These values indicate that the executed cable shape remains close to the planned goal in the tested quasi-static replay conditions.

The error values also reflect the geometry of the tasks. **Scenario 4** has the lowest RMSE, 1.60 mm, which is consistent with a smoother deformation and relatively simple endpoint motion. **Scenarios 7** and **9** have higher RMSE, 4.66 mm and 4.58 mm, respectively. These cases require stronger deformation and larger cumulative endpoint motion, so small effects from cable elasticity, contact response, and numerical settling have a larger influence on the final configuration. Figure **??** shows representative executions at the start, two intermediate stages, and the

Table 1. : Benchmark performance for the selected GCR scenarios.

Scenario	Success (%)	Time (s)	Fastest (s)	Iterations (10^3)	Raw Path Configurations	Refined Path Configurations
1	100.00	0.240 ± 0.143	0.010	2.08 ± 1.20	17 ± 4	5 ± 1
2	100.00	0.283 ± 0.197	0.018	1.75 ± 1.21	20 ± 4	5 ± 1
3	100.00	0.008 ± 0.003	0.002	0.07 ± 0.03	13 ± 2	3 ± 0
4	100.00	1.306 ± 0.871	0.124	6.98 ± 4.52	22 ± 5	5 ± 1
5	100.00	0.017 ± 0.005	0.005	0.09 ± 0.03	16 ± 3	7 ± 0
6	100.00	0.145 ± 0.061	0.014	0.95 ± 0.42	18 ± 4	7 ± 1
7	100.00	0.151 ± 0.066	0.020	1.36 ± 0.57	19 ± 4	7 ± 1
8	100.00	0.072 ± 0.033	0.008	0.65 ± 0.31	16 ± 3	6 ± 1
9	100.00	0.104 ± 0.059	0.004	0.72 ± 0.41	18 ± 3	5 ± 1
10	100.00	0.339 ± 0.139	0.054	2.60 ± 1.08	25 ± 4	9 ± 1

Table 2. : Robotic manipulation validation results.

Scenario	RMSE (mm)	$e_{\max}$ (mm)
1	2.20	3.41
2	2.22	3.72
3	4.32	6.49
4	1.60	2.31
5	1.87	2.70
6	2.08	3.32
7	4.66	7.81
8	4.13	6.28
9	4.58	7.32
10	2.83	4.73

final state. The manipulation-validation records are available at https://github.com/KaramAlmaghout/gcr.

5.5. Summary of Findings

The benchmark evaluation shows that the proposed planner solves all selected planar scenarios over 1000 trials per scenario, with runtime increasing as the routing task becomes more constrained. The shortcut refinement step reduces the number of retained configurations and therefore produces paths that are more compact for execution. The simulation-based robotic manipulation validation indicates that the GCR paths can guide the manipulators toward the target cable shape under the tested replay conditions.

These results support the use of GCR as an offline or supervisory planner for planar cable-routing tasks.

6. Discussion

The results show that the proposed GCR planner can generate feasible routing paths for a cable-like deformable linear object in two-dimensional cluttered environments. The main contribution of the method is that it plans in the space of complete cable configurations rather than only in the space of endpoint positions. This allows the planner to reason about the shape of the whole cable, including local segment deformation, bending, static collision avoidance, and transition feasibility between consecutive configurations.

A central aspect of the proposed formulation is the distinction between configuration validity and transition validity. A cable configuration may be collision-free when considered alone, but the deformation from one configuration to another may still sweep through an obstacle. For this reason, GCR validates each accepted edge using a swept-envelope condition. This makes the resulting path more suitable for execution than a sequence of independently valid cable shapes, because each neighboring pair of configurations is checked as a local deformation.

The ordered representation of the cable is also taken into account. Since the first and last cable points correspond to endpoint labels, changing the order of a cable configuration changes the point-to-point correspondence used during transition validation. The ordered edge-feasibility condition therefore ensures that the same endpoint correspondence used by the planner is preserved in the final path. This is relevant for dual-arm manipulation, where each endpoint is assigned to a specific end-effector and the endpoint correspondence affects execution feasibility.

The combination of APF-biased sampling and bidirectional tree search provides a practical balance between exploration and guidance. The APF component improves the quality of sampled configurations by moving them toward the opposite tree and away from obstacle regions. At the same time, the planner does not rely on APF descent alone; the bidirectional tree structure preserves global exploration and allows the search to recover from local geometric restrictions.

The projected elastic relaxation step contributes to the mechanical plausibility of the generated configurations. It corrects APF-biased samples by preserving local segment structure and discrete curvature while keeping the candidate close to the sampled direction. This relaxation is not intended to replace a full physical cable simulator. Instead, it provides a computationally efficient correction step that makes sampled configurations more consistent with the discrete cable model before they are validated and inserted into the tree.

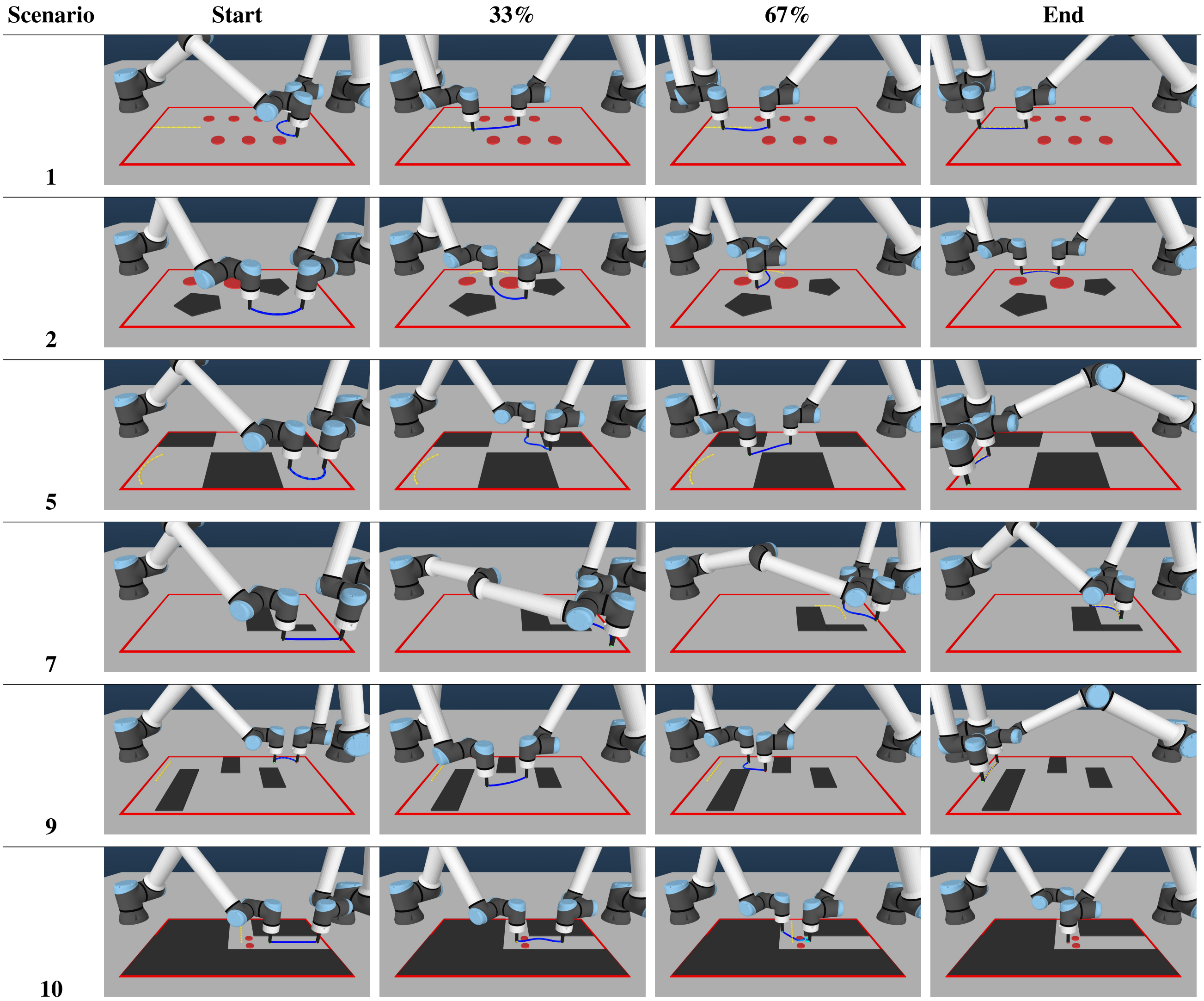


Figure 8: Execution snapshots for the validation scenarios. Each row shows the simulated cable manipulation at the start, two intermediate, and the end configurations.

The benchmark results indicate that the proposed pipeline is effective for the selected planar routing tasks. The planner found feasible paths in all tested scenarios, and the computation time remained suitable for offline or supervisory planning. More constrained scenarios required more iterations and longer planning time, which is expected because narrow passages and obstacle-dense regions reduce the number of admissible local transitions. The shortcut refinement step further reduced the number of retained configurations, producing compact paths that are easier to pass to an execution layer. This refinement also reduces unnecessary manipulation effort: redundant intermediate configurations are removed only when the direct transition remains feasible, and the deformation cost penalizes abrupt changes in segment vectors and discrete curvature. As a result, the final path does not merely connect the start and goal configurations; it also avoids unnecessary cable reshaping and provides a shorter sequence of deformation-feasible manipulation steps.

The robotic manipulation validation provides an additional indication that the planned paths can be used to generate endpoint trajectories for dual-arm manipulation. The robots in the simulation setup were able to follow the endpoint references and reproduce the planned target cable shapes with small final configuration errors. This result supports the use of GCR as a planning layer for generating deformation-feasible reference paths.

The present formulation defines a focused scope for the first implementation of GCR and also indicates several directions for further development. In this work, the planner is evaluated for two-dimensional planar routing with known static obstacles. This setting is relevant to quasi-planar manipulation tasks, but future extensions should consider three-dimensional cable routing, out-of-plane motion, torsion, and contact interactions. The current cable model is geometric and quasi-static: segment and curvature regularization are used to preserve locally plausible shapes, while forces, stress limits, frictional contacts, and dynamic effects are not explicitly simulated during planning. Incorporating such physical quantities would make the method applicable to a wider range of deformable-object manipulation tasks.

The swept-envelope test used in GCR provides an efficient geometric criterion for rejecting unsafe local transitions. Since it approximates the occupied region during a deformation, future work should study more accurate transition models, especially when the real cable motion differs from the assumed interpolation. The validation presented in this paper is limited to a finite set of planar benchmark scenarios and a simulation-based dual-arm setup. These experiments demonstrate the feasibility of the proposed planning pipeline within the tested conditions, while further evaluation is needed across more diverse obstacle layouts, cable stiffness values, robot platforms, and real experimental setups.

The proposed planner is most directly applicable to tasks where a flexible linear object must be routed through a cluttered but mainly planar workspace. Examples include robotic cable routing in electrical cabinet assembly, wire harness installation, routing of flexible tubes or hoses in manufacturing, and planning intermediate shapes for dual-arm manipulation of ropes, wires, or similar objects. In such applications, GCR can serve as a supervisory planner that produces a sequence of feasible cable configurations, while a lower-level controller tracks the corresponding endpoint motions and compensates for execution errors.

7. Conclusion

This paper presented GCR, a global planner for routing cable-like deformable linear objects in two-dimensional cluttered environments. The method represents the cable as an ordered polygonal chain and searches directly in the corresponding configuration space. This allows the planner to evaluate not only the feasibility of individual cable shapes, but also the feasibility of local deformations between consecutive configurations.

The proposed planning pipeline combines bidirectional tree search, APF-guided configuration sampling, projected elastic relaxation, ordered transition validation, and shortcut refinement. The benchmark results showed that the planner generated feasible paths in the tested planar scenarios, while the refinement stage reduced redundant intermediate configurations and avoided unnecessary deformation. The simulation-based manipulation validation further indicated that the resulting paths can be converted into endpoint references that reproduce the target cable shapes with small final errors under the tested conditions.

Overall, the results support the use of GCR as an offline or supervisory planning layer for planar cable-routing tasks. Future work will focus on extending the formulation to three-dimensional environments, incorporating richer physical models of cable deformation and contact, and evaluating the planner on real robotic systems.